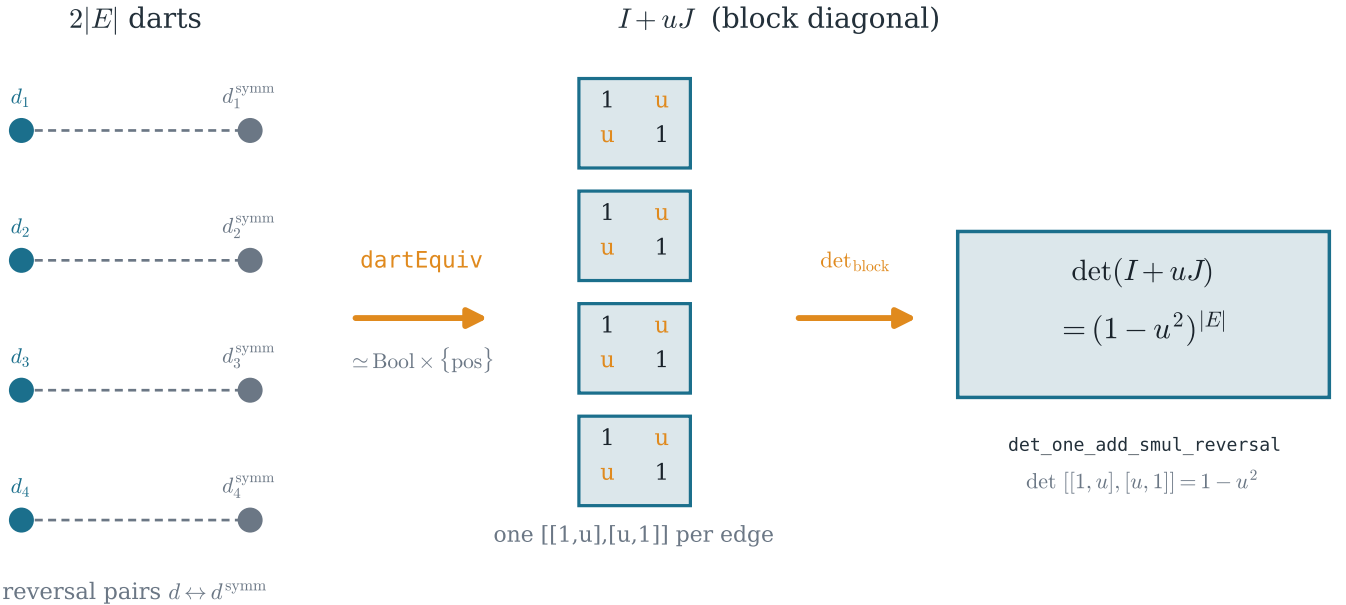Orientation reindex: J is $|E|$ disjoint swaps, so $I + uJ$ is block-diagonal and $\det(I + uJ) = (1 - u^2)^{|E|}$